Knowledge Tracing - Continued

Machine Learning for Behavioral Data

March 17, 2026

EPFL

IC

Today's Topic

Week	Lecture/Lab
1	Data Exploration
2	Regression
3	Classification + Model Evaluation
4	Evaluation + Knowledge Tracing
5	**Knowledge Tracing**
6	Time Series Prediction
7	Unsupervised Learning
8	Spring Break

Getting ready for today's lecture…

- Join us on SpeakUp

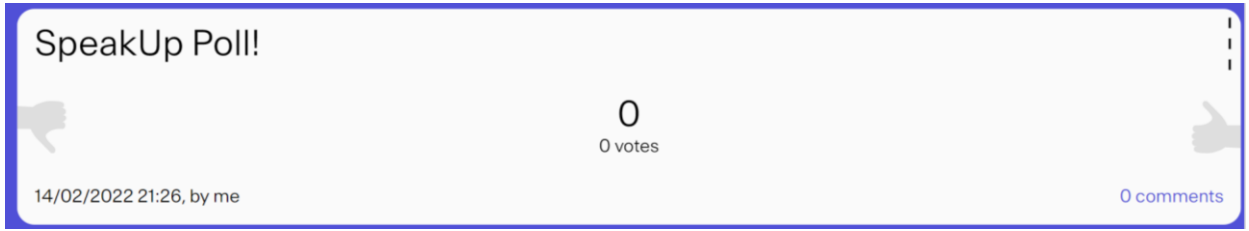
https://go.epfl.ch/mlbd-speakup

Short quiz about the past…

Mixed-effect models are useful when the samples in the data set are uncorrelated.

a) True

b) False



https://go.epfl.ch/mlbd-speakup

Short quiz about the past…

A data scientist builds a regression model to predict increasing student knowledge over the semester. They assume that students start at different levels of prior knowledge, but all have the same learning rate. Which of the following is true?

The planned regression model uses

a) a fixed intercept and a random slope.

b) a fixed intercept and a fixed slope.

c) a random intercept and a fixed slope.

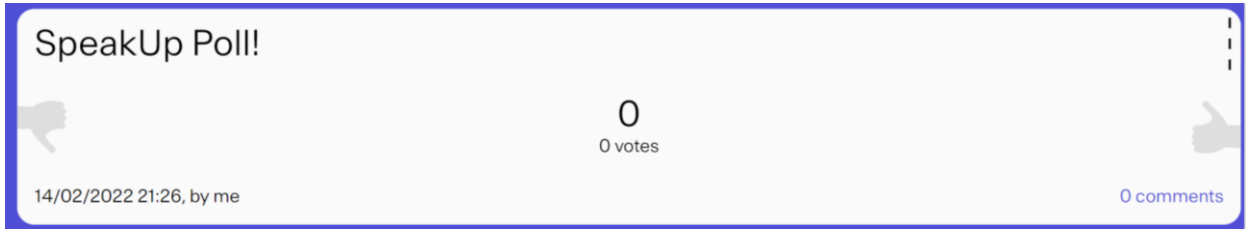
d) a random intercept and a random slope.

https://go.epfl.ch/mlbd-speakup

Short quiz about the past…

BKT does account for students guessing the correct answer.

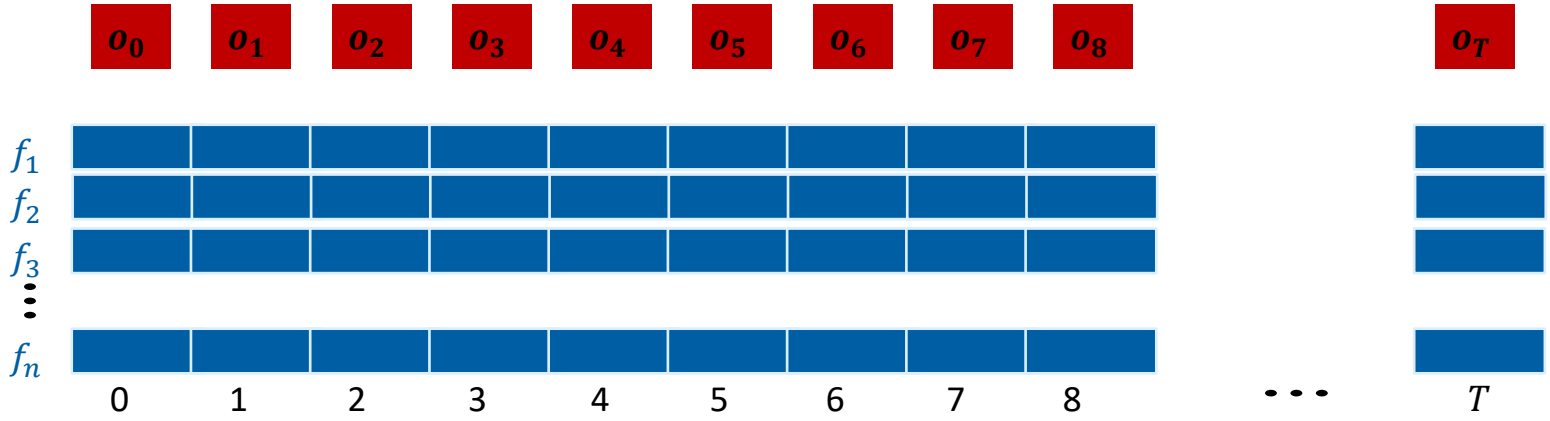
a) True
b) False



https://go.epfl.ch/mlbd-speakup

Time Series – Tracing Task

- Prediction of a variable in time step $t + 1$, based on time steps $0, \dots, t$



Tracing Knowledge – why is it useful?

- Is the student learning?
  - Measure what the student *knows* at a specific time *t*
  - More specifically: knowledge of the student about relevant knowledge components (skills)

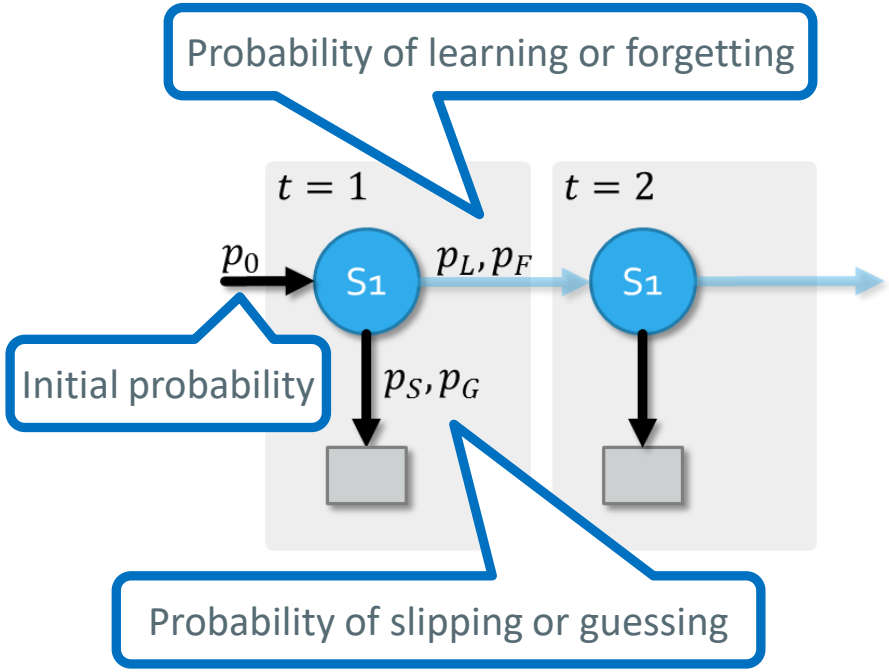
➡ Choose the next appropriate activity

➡ Know which activities support learning

Agenda

- **Bayesian Knowledge Tracing:**
  - **Inference - Recap**
  - **Training**
- Learning Curves
- Alternative models for knowledge tracing
  - AFM
  - PFA

Last Week: Bayesian Knowledge Tracing

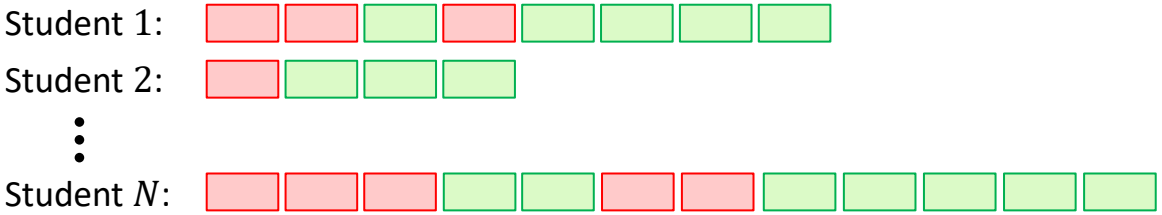


A student solves tasks i_{s_k} associated with a skill s_k. We want to predict

- $p(o_t | o_0, \dots, o_{t-1})$, the probability that the student will solve task i_{s_k} correctly at time step t
- $p(s_{k,t} | o_0, \dots, o_{t-1})$, the probability that the student has mastered skill s_k at time step t

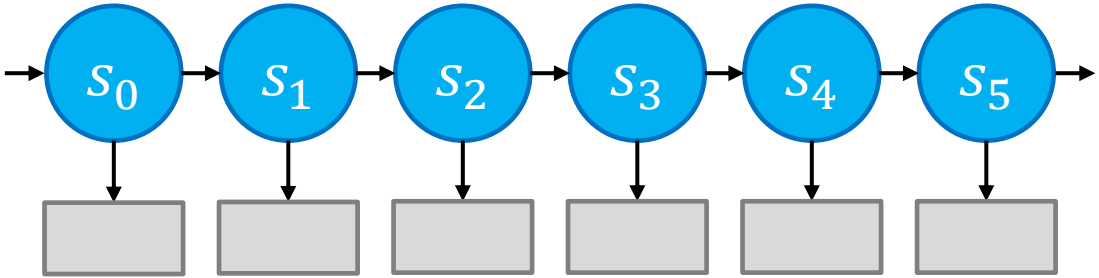
Two tasks need to be solved in practice

- Given sequences of observations $\boldsymbol{o} = [o_0, \dots, o_T]$ of N students, learn the parameters $\theta = \{p_0, p_L, p_F, p_S, p_G\}$ that maximize the likelihood of the observed data



Parameter Learning

Training a BKT model



$$\theta = \{p_0, p_L, p_F, p_S, p_G\}$$

Student l_0: $\mathbf{o}_{l_0} = [0,1,1]$

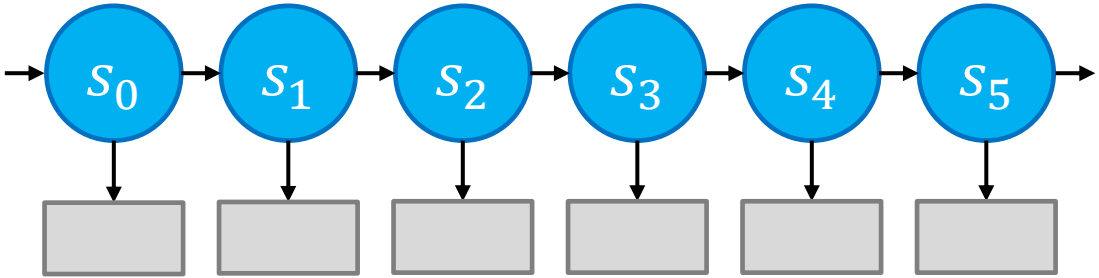
⋮

Student l_{N-1}: $\mathbf{o}_{l_{N-1}} = [1,0,1,1,1,0,0,1,1,1]$

Student l_N: $\mathbf{o}_{l_N} = [0,1,0,1]$

$$\max_{\theta} \; p\left(\boldsymbol{o}_{l_0}, \dots, \mathbf{o}_{l_{N-1}}, \mathbf{o}_{l_N}\right)$$

Training a BKT model



$$\theta = \{p_0, p_L, p_F, p_S, p_G\}$$

Student l_0: $\mathbf{o}_{l_0} = [0,1,1]$

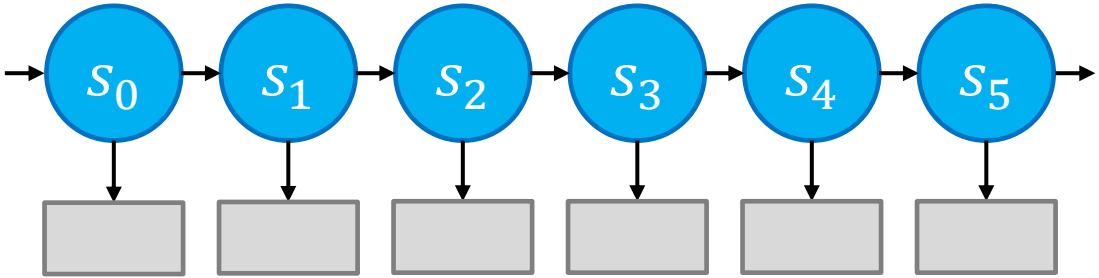
⋮

Student l_{N-1}: $\mathbf{o}_{l_{N-1}} = [1,0,1,1,1,0,0,1,1,1]$

Student l_N: $\mathbf{o}_{l_N} = [0,1,0,1]$

$$\max_{\theta} \prod_{i=1}^{N} p(\boldsymbol{o}_{l_i})$$

Training a BKT model



$$\theta = \{p_0, p_L, p_F, p_S, p_G\}$$

Student l_0: $p(\mathbf{o}_{l_0}) = \sum_{\boldsymbol{s}} p(\mathbf{o}_{l_0}, \boldsymbol{s})$

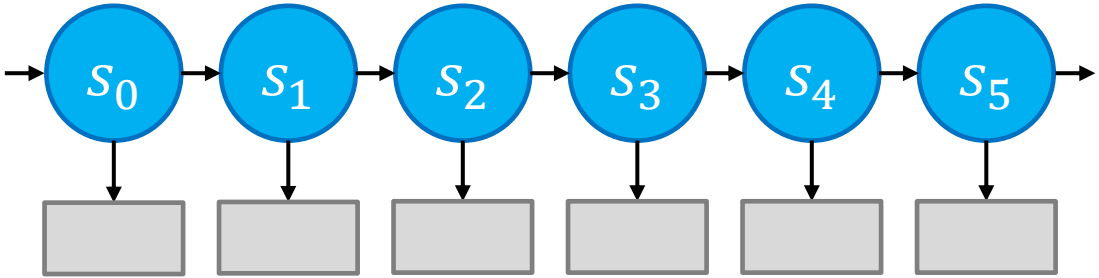
⋮

Student l_{N-1}: $p(\mathbf{o}_{l_0}) = \sum_{\boldsymbol{s}} p(\mathbf{o}_{l_0}, \boldsymbol{s})$

Student l_N: $p(\mathbf{o}_{l_0}) = \sum_{\boldsymbol{s}} p(\mathbf{o}_{l_0}, \boldsymbol{s})$

$$\max_{\theta} \prod_{i=1}^{N} \sum_{\boldsymbol{s}_{l_i}} p(\mathbf{o}_{l_i}, \boldsymbol{s}_{l_i})$$

Training a BKT model



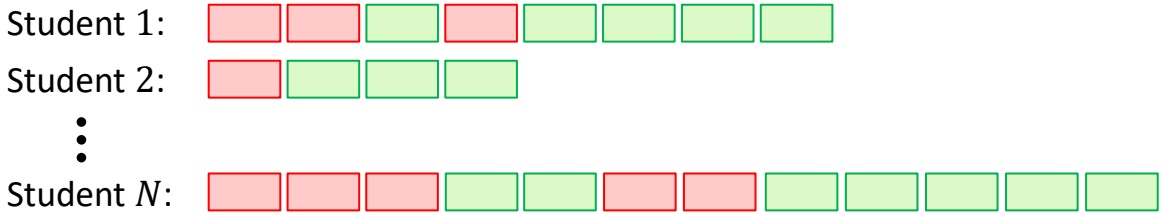
$$\theta = \{p_0, p_L, p_F, p_S, p_G\}$$

$$\max_{\theta} \prod_{i=1}^{N} \sum_{\boldsymbol{s}_{l_i}} p(\mathbf{o}_{l_i}, \boldsymbol{s}_{l_i}) \Rightarrow \min_{\theta} -\sum_{i=1}^{N} \log\left(\sum_{\boldsymbol{s}_{l_i}} p(\mathbf{o}_{l_i}, \boldsymbol{s}_{l_i})\right)$$

- Brute-Force Grid Search
- Expectation Maximization
- Gradient Descent
- Nelder-Mead Optimization

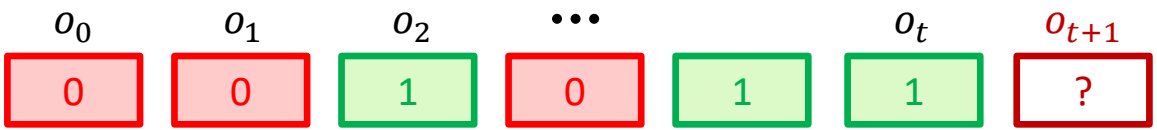
Two tasks need to be solved in practice

- Given sequences of observations $\boldsymbol{o} = [o_0, \dots, o_T]$ of N students, learn the parameters $\theta = \{p_0, p_L, p_F, p_S, p_G\}$ that maximize the likelihood of the observed data



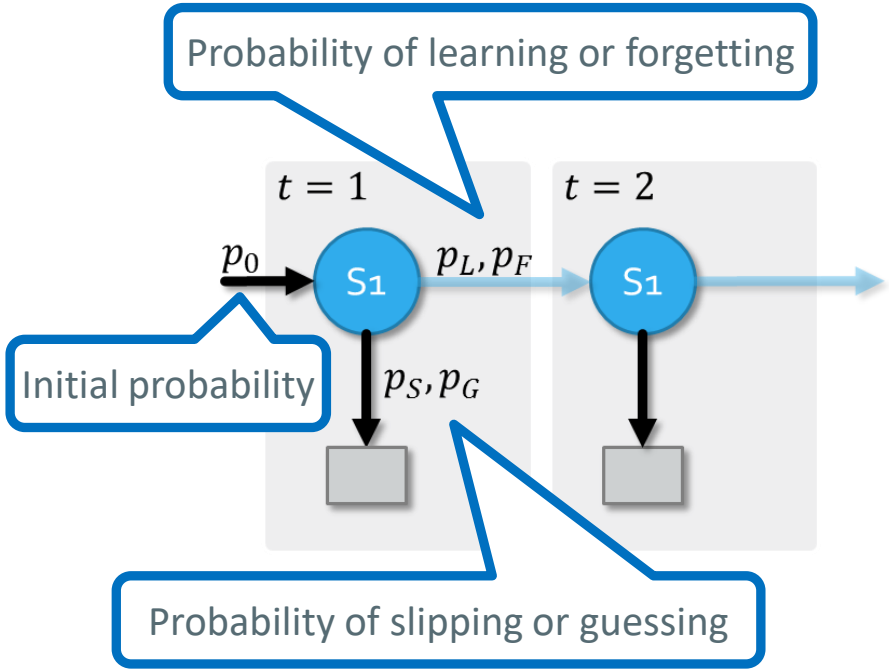
Parameter Learning

- Given a model with parameters $\theta = \{p_0, p_L, p_F, p_S, p_G\}$ and a sequence of observations $\boldsymbol{o} = [o_0, \dots, o_t]$ from a student s, predict o_{t+1}



Inference

BKT – What are we predicting?



A student solves tasks i_{s_k} associated with a skill s_k. We want to predict

- $p(o_t | o_0, \dots, o_{t-1})$, the probability that the student will solve task i_{s_k} correctly at time step t
- $p(s_{k,t} | o_0, \dots, o_{t-1})$, the probability that the student has mastered skill s_k at time step t

Inference in BKT models

$\boldsymbol{o_{t-1}} = [o_0, \dots, o_{t-1}]$

Equations for time $t = 0$:

Belief about latent state before observation

$$p(s_0 = 1) = p_0$$

Predicted observation at time t

$$p(o_0 = 1) = (1 - p_S) \cdot p_0 + p_G \cdot (1 - p_0)$$

$$p(o_0 = 0) = p_S \cdot p_0 + (1 - p_G) \cdot (1 - p_0)$$

Posterior: belief about latent state after observation

$$\left.\begin{aligned} p_{s_0|1} &= \frac{(1 - p_S) \cdot p_0}{(1 - p_S) \cdot p_0 + p_G \cdot (1 - p_0)} \\ p_{s_0|0} &= \frac{p_S \cdot p_0}{p_S \cdot p_0 + (1 - p_G) \cdot (1 - p_0)} \end{aligned}\right\} p_{s_0|o_0}$$

Equations for time steps $t = 1, \dots, T$:

$$p_{s_t|\boldsymbol{o_{t-1}}} = (1 - p_F) \cdot p_{s_{t-1}|\boldsymbol{o_{t-1}}} + p_L \cdot (1 - p_{s_{t-1}|\boldsymbol{o_{t-1}}})$$

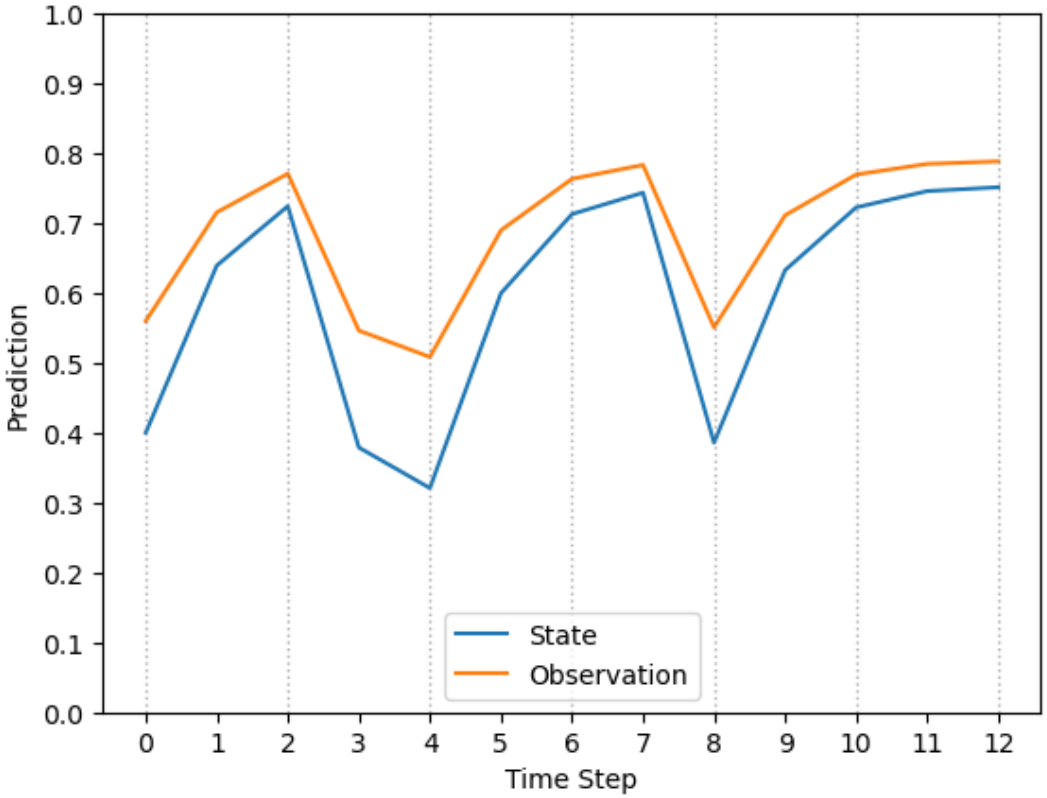
$$p(o_t = 1|\boldsymbol{o_{t-1}}) = (1 - p_S) \cdot p_{s_t|\boldsymbol{o_{t-1}}} + p_G \cdot (1 - p_{s_t|\boldsymbol{o_{t-1}}})$$

$$p(o_t = 0|\boldsymbol{o_{t-1}}) = p_S \cdot p_{s_t|\boldsymbol{o_{t-1}}} + (1 - p_G) \cdot (1 - p_{s_t|\boldsymbol{o_{t-1}}})$$

$$\left.\begin{aligned} p_{s_t|1,\boldsymbol{o_{t-1}}} &= \frac{(1 - p_S) \cdot p_{s_t|\boldsymbol{o_{t-1}}}}{(1 - p_S) \cdot p_{s_t|\boldsymbol{o_{t-1}}} + p_G \cdot (1 - p_{s_t|\boldsymbol{o_{t-1}}})} \\ p_{s_t|0,\boldsymbol{o_{t-1}}} &= \frac{p_S \cdot p_{s_t|\boldsymbol{o_{t-1}}}}{p_S \cdot p_{s_t|\boldsymbol{o_{t-1}}} + (1 - p_G) \cdot (1 - p_{s_t|\boldsymbol{o_{t-1}}})} \end{aligned}\right\} p_{s_t|\boldsymbol{o_t}}$$

[Corbett & Anderson, UMUAI 1994]

Your Turn - Inference



A student solves tasks i_{s_k} associated with a skill s_k. We predict

- **Orange**: $p(o_t | o_0, \dots, o_{t-1})$, the probability that the student will solve the task correctly at time step t
- **Blue**: $p(s_{k,t} | o_0, \dots, o_{t-1})$, the probability that the student has mastered skill s_k at time step t

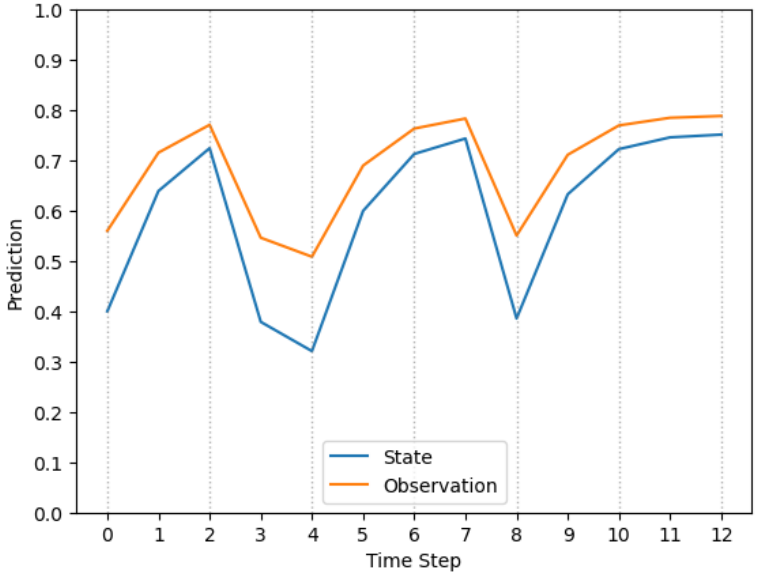
Your Turn - Inference

Which parameter combinations can produce the plot of predicted observations and knowledge state (see previous slide)? Explain your selection.

- ☐ $p_0 = 0.55, p_s = 0.05, p_g = 0.3, p_l = 0.3, p_f = 0.2, o = [1,1,0,0,1,1,1,0,1,1,1,1]$
- ☐ $p_0 = 0.4, p_s = 0.05, p_g = 0.3, p_l = 0.3, p_f = 0.2, o = [1,1,0,0,1,1,1,0,1,1,1,1]$
- ☐ $p_0 = 0.4, p_s = 0.5, p_g = 0.3, p_l = 0.3, p_f = 0.2, o = [1,1,0,0,1,1,1,0,1,1,1,1]$
- ☐ $p_0 = 0.4, p_s = 0.05, p_g = 0.3, p_l = 0.3, p_f = 0.2, o = [1,1,0,1,1,1,1,0,1,1,1,1]$

Your Turn - Inference

- ☐ $p_0 = 0.55, p_s = 0.05, p_g = 0.3, p_l = 0.3, p_f = 0.2, o = [1,1,0,0,1,1,1,0,1,1,1,1]$
- ☐ $p_0 = 0.4, p_s = 0.05, p_g = 0.3, p_l = 0.3, p_f = 0.2, o = [1,1,0,0,1,1,1,0,1,1,1,1]$
- ☐ $p_0 = 0.4, p_s = 0.5, p_g = 0.3, p_l = 0.3, p_f = 0.2, o = [1,1,0,0,1,1,1,0,1,1,1,1]$
- ☐ $p_0 = 0.4, p_s = 0.05, p_g = 0.3, p_l = 0.3, p_f = 0.2, o = [1,1,0,1,1,1,1,0,1,1,1,1]$



Assumptions behind BKT

- Knowledge can be divided into different skills
- Definition of skills is accurate/detailed enough
- Each task corresponds to a single skill (original)
- There is **no** connection between the skills
- Mastery can be achieved through practice
- There is no forgetting: $p_F = 0$ (original)

Agenda

- Bayesian Knowledge Tracing:
  - Inference - Recap
  - Training
- **Learning Curves**
- Alternative models for knowledge tracing
  - AFM
  - PFA

Tracing Knowledge – why is it useful?

- Is the student learning?
  - Measure what the student *knows* at a specific time *t*
  - More specifically: knowledge of the student about relevant knowledge components (skills)

➡ Choose the next appropriate activity

➡ Know which activities support learning

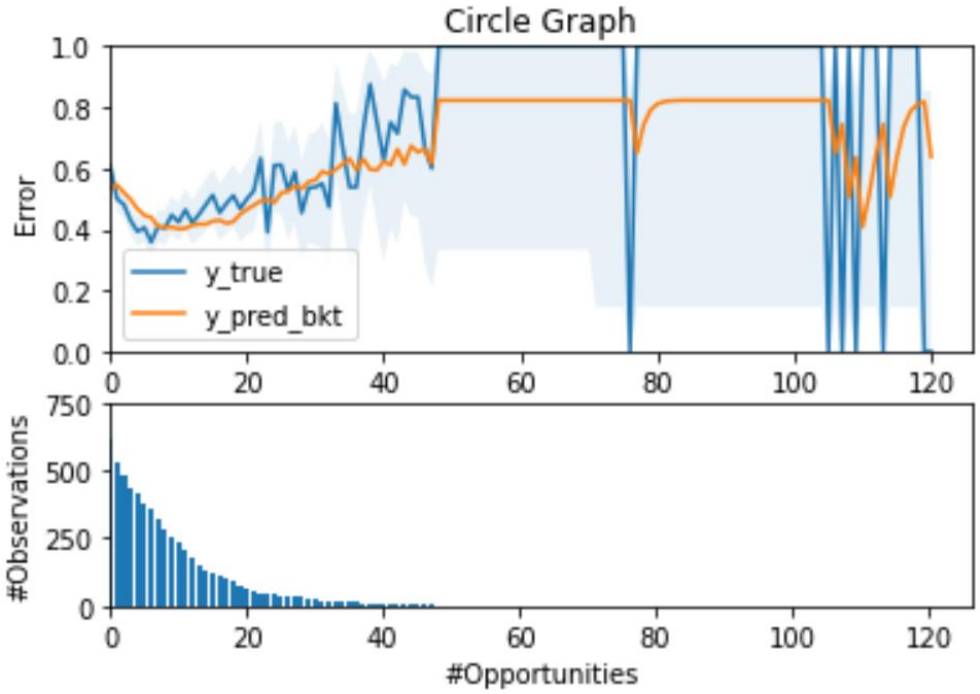
Building a learning curve for skill s

Student	Opportunity	y_true	y_pred
0	0	0	0.3
0	1	0	0.5
0	2	1	0.7
0	3	1	0.9
1	0	0	0.3
1	1	1	0.5
2	0	0	0.3
2	1	1	0.5
2	2	1	0.7
3	0	1	0.3
3	1	0	0.7
3	2	1	0.5
3	3	1	0.9

Use Case

- ASSISTments is a free tool for assigning and assessing math problems and homework
- All math problems (tasks/items) are associated to a specific skill/knowledge component
- 4,151 middle-school students
- 525,534 observations

What could this curve indicate?



https://go.epfl.ch/mlbd-speakup

Your Turn – Learning Curves

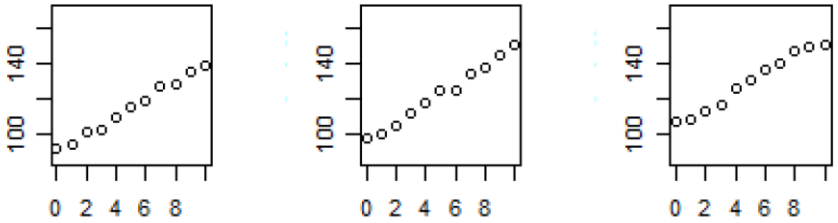
- In the student notebook, you have:
  - BKT model trained on all skills and students
  - List of available skills
  - Function for plotting learning curves and student numbers for a specific skill
- Your task:
  - Pick 1-2 skills, generate the learning curves for them, and interpret them
  - Send us your plots and interpretations

Agenda

- Bayesian Knowledge Tracing:
  - Inference - Recap
  - Training
- Learning Curves
- **Alternative models for knowledge tracing**
  - **AFM**
  - **PFA**

Generalized Linear Mixed Effects Models revisited

- Example: strength gain by weight training
  - Each person has individual starting strength



$$y_n = \beta_0 + u_n + \beta_1 x_{n,1}$$

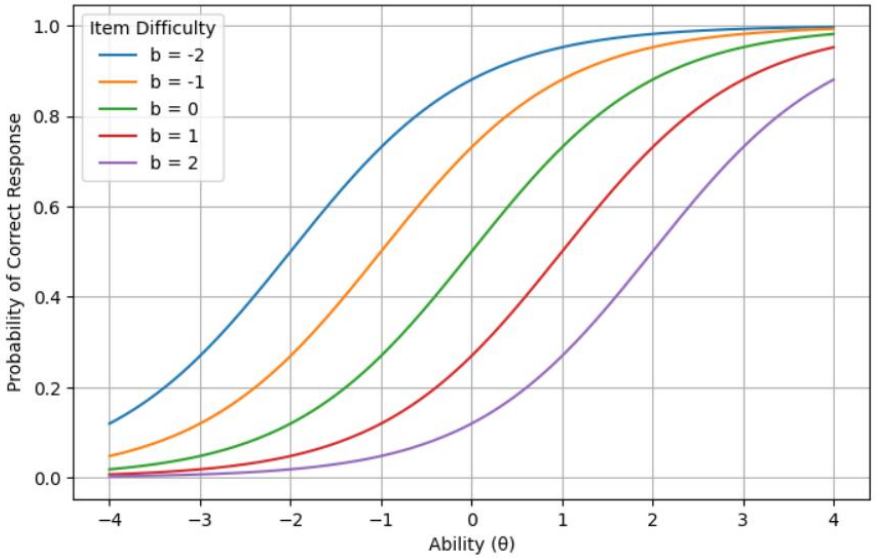
"Fixed" Effects

"Random" Effect

Rasch Model

$$\log\left(\frac{p_{i,n}}{1-p_{i,n}}\right) = \theta_n - b_i$$

Probability that student n will solve item i correctly.



θ_n: student ability

b_i: difficulty of item i

Additive Factors Model (AFM)

$$p_{n,i} = \frac{1}{1 + e^{-\pi_{n,i}}}$$

Probability that student n will solve task i correctly.

Additive Factors Model (AFM)

$$p_{n,i} = \frac{1}{1 + e^{-\pi_{n,i}}}$$

$$\pi_{n,i} = \theta_n + \sum_k q_{i,k} \cdot (\beta_k + \gamma_k \cdot T_{n,k})$$

[Cen et al., ITS 2006; Cen et al., ITS 2008]

Additive Factors Model (AFM)

$$p_{n,i} = \frac{1}{1 + e^{-\pi_{n,i}}}$$

$$\pi_{n,i} = \theta_n + \sum_k q_{i,k} \cdot (\beta_k + \gamma_k \cdot T_{n,k})$$

Student proficiency

[Cen et al., ITS 2006; Cen et al., ITS 2008]

Additive Factors Model (AFM)

$$p_{n,i} = \frac{1}{1 + e^{-\pi_{n,i}}}$$

$$\pi_{n,i} = \theta_n + \sum_k q_{i,k} \cdot (\beta_k + \gamma_k \cdot T_{n,k})$$

Student proficiency

$q_{ik} = 1$, if item *i* uses skill *k*

[Cen et al., ITS 2006; Cen et al., ITS 2008]

Additive Factors Model (AFM)

$$p_{n,i} = \frac{1}{1 + e^{-\pi_{n,i}}}$$

$$\pi_{n,i} = \theta_n + \sum_k q_{i,k} \cdot (\beta_k + \gamma_k \cdot T_{n,k})$$

Student proficiency

Difficulty of skill *k*

$q_{ik} = 1$, if item *i* uses skill *k*

[Cen et al., ITS 2006; Cen et al., ITS 2008]

Additive Factors Model (AFM)

$$p_{n,i} = \frac{1}{1 + e^{-\pi_{n,i}}}$$

$$\pi_{n,i} = \theta_n + \sum_k q_{i,k} \cdot (\beta_k + \gamma_k \cdot T_{n,k})$$

Student proficiency

$q_{ik} = 1$, if item *i* uses skill *k*

Difficulty of skill *k*

Learning rate at skill *k*

Number of practice opportunities student *n* had at skill *k*

[Cen et al., ITS 2006; Cen et al., ITS 2008]

AFM - Assumptions

- Students may initially know more or less
- Students learn at the same rate
- Some skills are more likely to initially be known
- Some skills are easier to learn than others
- Students learn with each practice opportunity
- Each item belongs to one or more skills

Performance Factors Analysis (PFA)

$$\pi_{n,i} = \theta_n + \sum_k q_{i,k} \cdot (\beta_k + \boxed{\gamma_k \cdot T_{n,k}})$$

[Pavlik et al., AIED 2009]

Performance Factors Analysis (PFA)

$$\pi_{n,i} = \theta_n + \sum_k q_{i,k} \cdot (\beta_k + \boxed{\gamma_k \cdot s_{n,k} + \rho_k \cdot f_{n,k}})$$

Number of prior **successes** student *n* had at skill *k*

Number of prior **failures** student *n* had at skill *k*

[Pavlik et al., AIED 2009]

PFA - Assumptions

- Students may initially know more or less
- Students learn at the same rate
- Some skills are more likely to initially be known
- Some skills are easier to learn than others
- Students learning rate differs for correct and wrong practice opportunities
- Each item belongs to one or more skills

Your Turn: Comparing Models

- We have evaluated AFM, PFA, and BKT on a subset of six skills. Your task:
  - Visualize the overall RMSE and AUC of the models such that it can easily be compared
  - Discuss the obtained results

Summary – Why tracing knowledge?

- Is the student learning?
  - Measure what the student *knows* at a specific time *t*
  - More specifically: knowledge of the student about relevant knowledge components (skills)

➡ Choose the next appropriate activity

➡ Know which activities support learning